

Torque margin for x-axis:

$$\Sigma \tau = \tau_{\text{motor}} - \tau_{\text{resisting}}:$$

$$\tau_{\text{motor}} = 0.42 \text{ N}\cdot\text{m}$$

$$\Sigma F_{\text{resisting}} = F_a + f + F_{\text{cutting}}$$



wood
spine

cutting
~40N

x-axis
contains
two motors
so half
re-load

1 kg: PETG

1.2 kg
SFUI204 x2

$$m \cdot a + m \cdot g + 40 \Rightarrow 41.5/2$$

$$100 \text{ mm/s}^2 \Rightarrow 0.1 \text{ m/s}$$

1.5 kg
SFUI605

$$\Rightarrow \approx 7.40 \text{ kg}$$

0.28 kg Nema
17

1.25 kg Nema 23

0.45 kg spindles

$$20.73 \text{ N}$$

convert to torque
4mm efficiency
0.4

$$\text{Derivation } \tau = \Sigma F \cdot L (2 \text{ m})$$

$$\tau = 0.1467 \text{ N}\cdot\text{m}$$

$$\therefore \Sigma \tau = \text{torque margin}$$

$$\approx 0.27 \text{ N}\cdot\text{m}$$

Load torque:

$$W_{\text{in}} = \tau \theta \text{ for one revolution}$$

$$W_{\text{out}} = Fd \text{ or } F \cdot L \text{ lead (linear distance the ball nut travels)}$$

η : efficiency: how much energy is maintained

$$= W_{out} / W_{in} = F \cdot L / (T \cdot 2\pi)$$

Solving for T gets us:

$$T = F \cdot L / (2\pi \cdot \eta)$$

That's how to get Torque Force for a lead screw

Torque Margin for y-axis:

$$\sum T = T_{motor} - T_{resisting}$$

$$F_{resisting} = F_a + f + F_{cutting} \Rightarrow$$

$$m \cdot a + \mu \cdot m \cdot g + 40N$$

0.60 kg $\leftarrow m$
from PE46

1.5 kg from SPU1605
 $= 3.4$

0.30 kg Nema 17

1 kg spindle

$$= 40.3675 N$$

$$T = \frac{40.3675 \cdot 0.04}{2\pi \cdot 0.9}$$

$$= 0.29 N \cdot m$$

$$\sum T = 0.42 N \cdot m - 0.29 N \cdot m = 0.134 N \cdot m$$

Torque Margin for Z-axis: 2.4 Nm

$$\sum T = \tau_{\text{motor}} - \tau_{\text{resisting}}$$

now gravity is acting directly on rotation

$$\tau_{\text{resisting}} = F_a + F_g + F_{\text{cutting}} = ma + mg + 40 \text{ N}$$

$$m = 100 \text{ mm/s}^2$$

$$0.3 \text{ kg} \text{ or } 0.01 \text{ m/s}^2$$

$$\text{PETB} = 1.3 \text{ kg}$$

1 kg spindle

$$1.3 \cdot 0.01 + 1.3 \cdot 9.81 + 40 = 52.766$$

$$\tau_{\text{resisting}} = F \cdot 0.05 / (2\pi \cdot 0.9) = 0.38 \text{ N}\cdot\text{m}$$

now lead is 5mm

$$\sum T = 2.4 - 0.38 \text{ N}\cdot\text{m}$$

$$\sum T \approx 2 \text{ N}\cdot\text{m}$$